

FRC
Communications
User Manual
For MADEE DeviceNet™ Series

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MADEE

Communication Specifications for DeviceNetTM

Digital Flow Ratio Controller

MADEE

DeviceNetTM Series



MKP Co., Ltd.

Designed by MKP

Revision History

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1. Introduction

Thank you for purchasing one of our MADEE Flow Ratio Controllers. It is recommended that this manual be thoroughly read and understood prior to installation and operation of this product.

2. Caution for use

- ① Do not use the FRC in locations exposed to high temperature, high humidity or vibration. The maximum storage temperature of the FRC is 65°C.
- ② Make sure your gas type and flow rate match the specification of this FRC.
- ③ Use of a gas different from the gas specified for the FRC may cause large deviations in the flow rate. The FRC cannot control the flow rate of some gas types.
- ④ Make sure the installation direction (gas flow direction) is correct. Connecting the FRC backwards may cause it to malfunction.
- ⑤ There must not be any gas leaks in the plumbing, especially when using gases at very low concentrations.
- ⑥ Use of gas at excessively high pressures may cause faulty operation of the FRC. Observe the specified operating pressure range for the FRC.
- ⑦ If the FRC is placed under pressure at a level that exceeds its withstanding pressure, it may be damaged.
- ⑧ Use of the FRC at operating temperatures over the specified range may cause faulty FRC operation. Use the FRC only within the range of 5 to 50°C.
- ⑨ Do not use contaminated gas. The FRC may become clogged or malfunction.
- ⑩ If you need to use highly reactive gas, you must first thoroughly purge the plumbing and the FRC itself.
- ⑪ Any residual oxygen or moisture left inside due to insufficient purging can cause a chemical reaction, which can damage the FRC.
- ⑫ The FRC cannot completely stop the flow of gas itself. If you need to shut off the gas completely, use a separate shut off valve.
- ⑬ Notice that our standard conditions for calibrated flow rates are 0°C and 1 ATM of pressure(SCCM, SLM).
- ⑭ Make sure the +24VDC power supply is providing the correct voltage, polarity, and capacity.
- ⑮ Reversed polarity may damage the unit.

3. Connector

1) Connector type

- Connector used : DeviceNet™ Male Micro Style Connector
- Vendor : DDK
- Part Number : M12F305AMA

- P C B Contacts
- Rear-mount configuration
- Applicable wire size : AWG22 max.

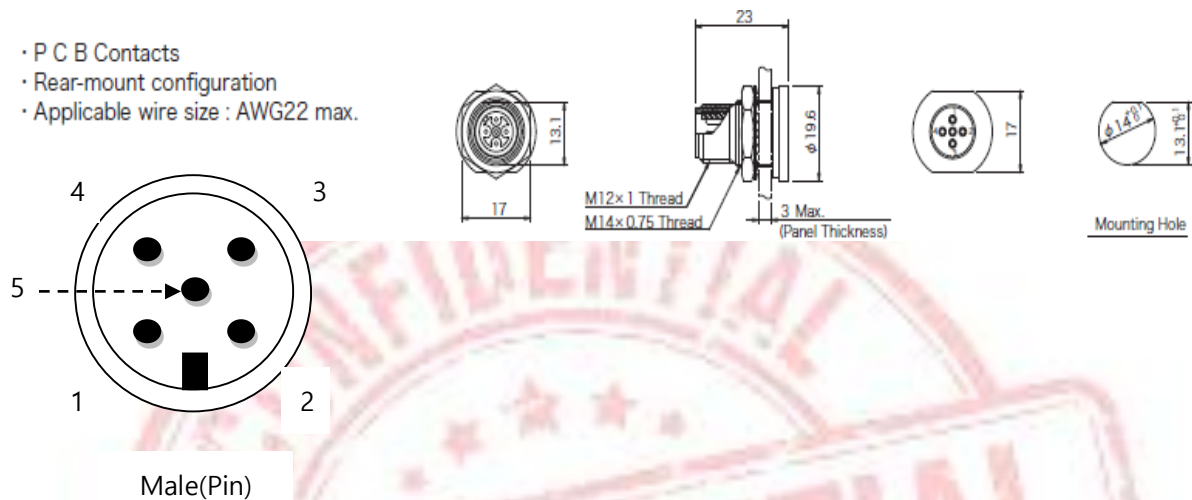


Fig 1. D-net Connector

Please refer product manual for more detail information.

2) Pin Assign

Table 3-1. Pin Assign of Connector

Pin No.	Signal Name	Color	Meaning
1	Drain	Bare	Shield
2	V+	Red	+24V DeviceNet Power Supply
3	V-	Black	Reference Potential DeviceNet Power Supply
4	CAN_H	White	CAN High-Signal
5	CAN_L	Blue	CAN Low-Signal

4. DeviceNet Hardware Settings

This paragraph describes the names, functions, and operation of each of the LEDs and rotary switches on the FRC.

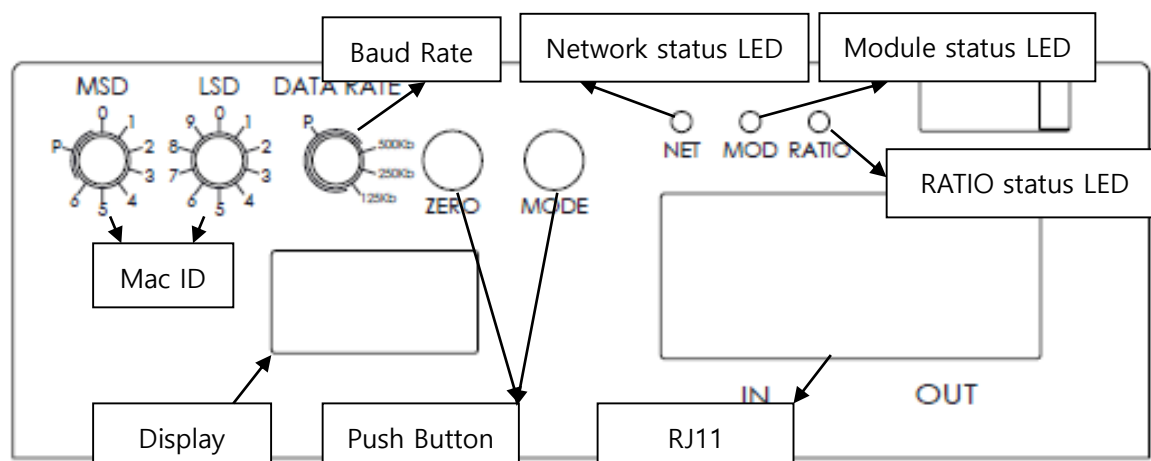


Fig 2. LEDs and Rotary switches on the top of the FRC

4-1. LED

The FRC is equipped with three, 2-color(green/red) LEDs to indicate the module status and network status and RATIO status.

The FRC controls these LEDs according to the module communication status.

The FRC follows the DeviceNet specifications in controlling the LEDs.

a. Display of the LEDs

Table 4-1. LED Operating

Display type	LED	Status	Lighting condition
Module status	OFF	No Power	The power is off.
	Light Green	Operating Normally	The power is on.
	Blink Green	Module Check	Self testing
	Light Red	Self testing Failure	Please, check the module.
Network status	OFF	Off line	Off line. Please check Mac ID.
	Light Green	On line/connected	Completed connection.
	Blink Green	On line/no connection	Module is on line but has no connections.
	Light Red	Communication error	*.Duplicate_Mac_ID test error *.Bus Off occurred
	Blink Red	Connection time out	
Ratio status	OFF	Off Control	No Ratio control
	Blink Red	In progress	Ratio control in progress
	Light Green	Stabilize	Stable Ratio control

4-2. Rotary Switch

The FRC is equipped with a switch for setting the communication speed(Baud rate) and a switch for setting the MacID.

The FRC reads the settings of the rotary switches when the power is turned on and sets the MacID internally.

1) Baud Rate Switch

Rotary Switch	Baud Rate
0	125kbps
1	250kbps
2	500kbps

2) Mac ID Switch

Rotary Switch(MSD)	Rotary Switch(LSD)
Upper Digit(x10)	Lower Digit(x1)

Mac ID = Upper Digit x 10 + Lower Digit

The MacID must be between 00 and 63(decimal).

4-3. Push Button

The FRC is equipped with a push button for setting the display value.

The FRC displays flow value when the power is turned on and be changed according to push button count.

Push Button Count	Display value
0	Actual Flow
1	Unit
2	Pressure
3	Temperature
4	Mac ID(The current set value)
5	Baud rate(The current set value)

4-4. RJ11(RS485)

Connect master and slave using RJ11.

The ratio setting can be output / set using a RS485 Communication.

However, if the ratio control is performed in Device Net Master, the ratio control cannot be controlled externally and only the ratio output can be checked.

5. Connection example

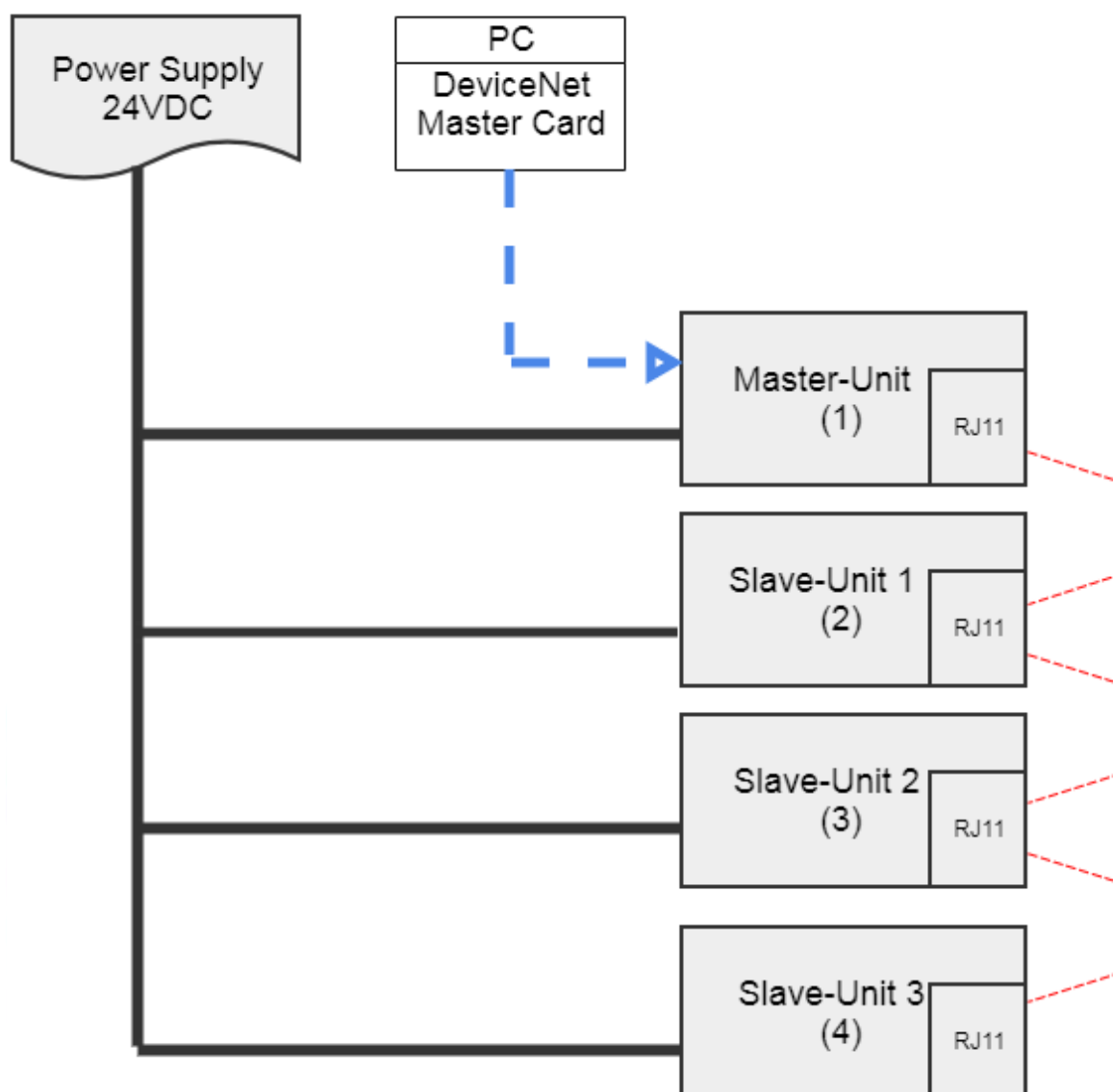


Fig 3. Connection example

6. DeviceNet Communication

6-1. Object Module

1) FRC device object module

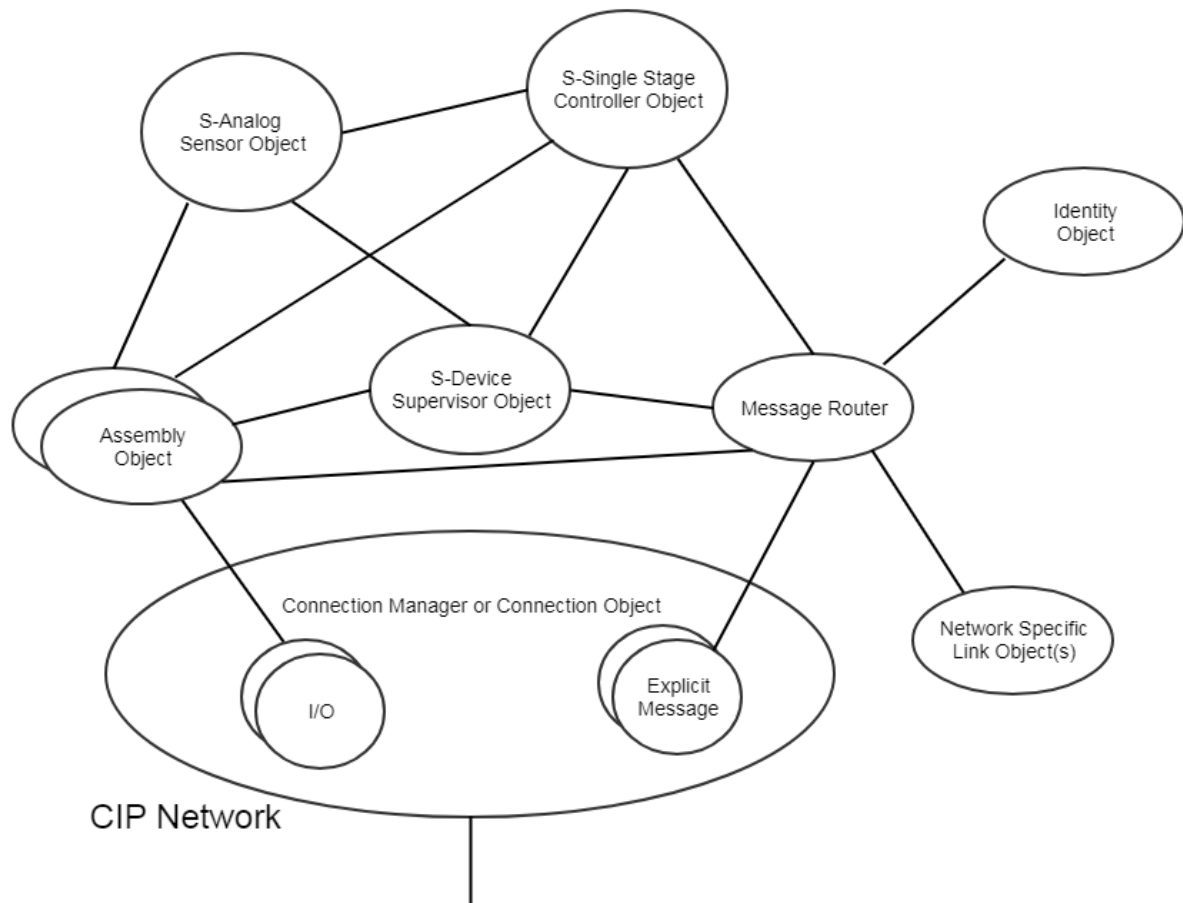


Fig 4. Object Model

2) Vendor Specific object classes

Table 6-1. Support List of Object

Object Class ID	Name	Optional/Required	Number of Instances
01 (0x01)	Identity	Required	1
02 (0x02)	Router	Required	1
03 (0x03)	DeviceNet	Required	1
04 (0x04)	Assembly	Required	At least 1 Input and 1 Output
05 (0x05)	Connection	Required	1
48 (0x30)	S-Device Supervisor	Required	1
49 (0x31)	S-Analog Sensor	Required	3
51 (0x33)	S-Single Stage Controller	Required	1

3) Supported Messages

Table 6-2. Support List of Messages

Message Group	Message ID	Type
2	03 hex	Slave's Explicit Response Messages
	04 hex	Master's Explicit Request Messages
	05 hex	Master's I/O Poll Command Messages
	06 hex	Unconnected Explicit Request Messages
	07 hex	Duplicate MAC ID Check Message

4) DeviceNet data types

Table 6-3. List of DeviceNet data types

Key Word	Description
BOOL	Boolean(0=False; 1=True)
INT	Integer(-32768 to 32767)
UINT	Unsigned integer(0 to 65536)
BYTE	Bit string(8 bits)
USINT	Unsigned short integer(0 to 255)
SHORT STRING	Character string(1 byte per character, 1byte length indicator)
REAL	Floating point(As per IEEE754)
UDINT	Unsigned long integer(0 to $2^{64}-1$)

6-2. Description of each Object

1) Identity Object(Class ID : 0x01)

- a. Instance : 1
- b. Attributes

Table 6-4. Identity Object Instance1 Attributes

Attr ID	Need for mounting	Access Rule	Name	Data type	Value
1	Required	Get	Vendor ID	UINT	1213
2	Required	Get	Device Type	UINT	0x0064(FRC)
3	Required	Get	Product Code	UINT	0x1010(MADEE Series)
4	Required	Get	Revision	STRUCT	1.0
			Major Revision	USINT	0x01
			Minor Revision	USINT	0x00
5	Required	Get	Status	WORD	0x0001
6	Required	Get	Serial Number	UDINT	
7	Required	Get	Product Name	SHORT STRING	" MADEE Series"
8		Get	State	USINT	

c. Services

Table 6-5. Identity Object Services

Service Code(HEX)	Need in implementation		Name
	Class	Instance	
0E	n/a	Required	Get_Attribute_Single
05	n/a	Required	Reset

2) Message Router Object(Class ID : 0x02)

- a. Instance : No
- b. Attribute : No
- c. Service : No

3) DeviceNet Object(Class ID : 0x03)

- a. Instance : 0 (Class)
- b. Attribute

Table 6-6. DeviceNet Object Instance0 Attributes

Attr ID	Need for mounting	Access Rule	Name	Data type	Vlaue
1	Required	Get	Revision	UINT	0x0002

- c. Service

Table 6-7. DeviceNet Object Instance0 Services

Service Code(HEX)	Need in implementation		Name
	Class	Instance	
0E	Optional	n/a	Get_Attribute_Single

d. Instance : 1

e. Attributes

Table 6-8. DeviceNet Object Instance1 Attributes

Attr ID	Need for mounting	Access Rule	Name	Data type	Value
1	Optional	Get	MAC ID	USINT	Range 0~63
2	Optional	Get	Baud Rate	USINT	Range 0~2 00 = 125kbps 01 = 250kbps 02 = 500kbps
3	Optional	Get	BOI	BOOL	Bus off Interrupt
4	Optional	Get	Bus-off Counter	USINT	Range 0~255
5	Optional	Get	Allocation Information	STRUCT Of :	DeviceNet Specifications
			Allocation Choice Byte	BYTE	0x00(default)
			Master's Mac ID	USINT	0xff(default)
6	Conditional	Get	MAC ID Switch Changed	BOOL	0= No Change 1= Change since last Reset or PowerUP
7	Conditional	Get	BaudRate Switch Changed	BOOL	0= No Change 1= Change since last Reset or PowerUP
8	Conditional	Get	MAC ID Switch Value	USINT	Range 0 ~ 99
9	Conditional	Get	BaudRate Switch Value	USINT	Range 0 ~ 9

f. Services

Table 6-9. DeviceNet Object Instance1 Services

Service Code(HEX)	Need in implementation		Name
	Class	Instance	
0E	Optional	Optional	Get_Attribute_Single
10	n/a	Optional	Set_Attribute_Single

4) Assembly Object(Class ID : 0x04)

The assembly instance of the I/O data can be selected by changing the procedure for Instance 2 of the connected object, or by the consumed connection path when all of the below are satisfied:

- After establishing Exp Msg data.
- Before establishing I/O Msg data.
- The FRC device status is idle
- N is the total number of Mater + Slave Units.

a. Supported Unit

Supported Unit	Detail
2	Use 2 Chanel
3	Use 3 Chanel
4	Use 4 Chanel
5	Use 5 Chanel
6	Use 6 Chanel

b. Input

Table 6-10. List of Assembly Object Instance Unit N

Instance ID	Byte Size	Detail
1	1+3*N	Control Mode + Override + Ratio Output
2	2+5*N	Status + Control Mode + Override + Ratio Output + Flow Output
3	2+3*N	Status + Control Mode + Override + Ratio Output ...
4	2+5*N	Status + Control Mode + Override + Ratio Output + Ratio Input

Table 6-11. Assembly Object Instance Input Detail of Total Unit N

Instance ID	Data type	Size	Byte	Data Detail
1	INT	1+3*N	0	Control Mode
			1	Master Unit Override
			2	Master Unit Ratio Output (low byte)
			3	Master Unit Ratio Output (high byte)
			...	...
			...	...
			...	...
			3*N -2	Slave N-1 Unit Override
			3*N -1	Slave N-1 Unit Ratio Output (low byte)
			3*N	Slave N-1 Unit Ratio Output (high byte)

2	INT	$2+5*N$	0	Status
			1	Control Mode
			2	Master Unit Override
			3	Master Unit Ratio Output (low byte)
			4	Master Unit Ratio Output (high byte)
			5	Master Unit Flow Output (low byte)
			6	Master Unit Flow Output (high byte)
			...	...
			$5*N-3$	Slave N Unit Override
			$5*N-2$	Slave N Unit Ratio Output (low byte)
			$5*N-1$	Slave N Unit Ratio Output (high byte)
			$5*N$	Slave N Unit Flow Output (low byte)
			$5*N+1$	Slave N Unit Flow Output (high byte)
3	INT	$2+3*N$	0	Status
			1	Control Mode
			2	Master Unit Override
			3	Master Unit Ratio Output (low byte)
			4	Master Unit Ratio Output (high byte)
			...	...
			$3*N-1$	Slave N-1 Unit Override
			$3*N$	Slave N-1 Unit Ratio Output (low byte)
			$3*N+1$	Slave N-1 Unit Ratio Output (high byte)
4	INT	$2+5*N$	0	Status
			1	Control Mode
			2	Master Unit Override
			3	Master Unit Ratio Output (low byte)
			4	Master Unit Ratio Output (high byte)
			5	Master Unit Flow Output (low byte)
			6	Master Unit Flow Output (high byte)
			...	...
			...	...
			...	...
			$5*N-3$	Slave N Unit Override
			$5*N-2$	Slave N Unit Ratio Output (low byte)
			$5*N-1$	Slave N Unit Ratio Output (high byte)
			$5*N$	Slave N Unit Flow Output (low byte)
			$5*N+1$	Slave N Unit Flow Output (high byte)

c. Output

Table 6-12. Assembly Object Output Data Mapping List

Instance ID	Access Rule	Name	Data type
7	Output	Control Mode, Ratio Input	INT

Table 6-13. Assembly Object Instance7 Detail

Instance ID	Data type	Size	Byte	Data Detail
7	INT	2N+1	0	Control Mode
			1	Master Unit Ratio Input (low byte)
			2	Master Unit Ratio Input (high byte)
			...	...
			...	...
			2N-1	Slave N-1 Unit Ratio Input (low byte)
			2N	Slave N-1 Unit Ratio Input (high byte)

d. I/O Assembly Data Mapping

Table 6-14. Assembly Object Data Mapping List

DataName	Class		Instance ID	Attribute		
	Name	ID		Name	ID	Type
Status	S-Device Supervisor	0x30	1	Exception Status	12	USINT
Control Mode	S-Single Stage Controller	0x31	1	Control Mode	5	USINT
Override	None					
Ratio Output	S-Single Stage Controller	0x33	1	Ratio Output	7	INT
Ratio Input	S-Single Stage Controller	0x33	1	Ratio Input	6	INT
Flow Output	S-Analog Sensor	0x31	1~n	Indicated Flow	6	INT

e. Control Mode Data Mapping

Control Mode Data	Detail
0x00	Control Mode
0x01	Full Close Mode
0x02	Full Open Mode
0xAA	AA Mode (equal control)

f. Override Data Mapping

Override Data	Detail
0x00	Control Mode
0x01	Close Mode
0x02	Open Mode
0xA0	Reminder Mode

g. Ratio Output, Ratio Input, Flow Output Value Range

Data	Detail
RatioOutput	Data MIN,MAX Range : 0x0000 ~ 0x8000 (0% ~ 100%)
RatioInput	Data MIN,MAX Range : 0x0000 ~ 0x8000 (0% ~ 100%)
FlowOutput	Data MIN,MAX Range : 0x0000 ~ 0x6000 (0% ~ 100%)

5) Connection Object(Class ID : 0x05)

a. Instance : 1 (Explicit Message)

b. Attributes

Table 6-15. Connection Object Instance1 Attributes

Attr ID	Access Rule	Name	Data type	Value	
1	Get	State	USINT	0x00	Non-existent
				0x01	Configuring
				0x02	Waiting for Connection ID
				0x03	Established
				0x04	Timed Out
				0x05	Deferred Delete
2	Get	Instance type	USINT	0x00(Explicit Message)	
3	Get	Transport Class Trigger	BYTE	83HEX(Server, Class 3)	
4	Get	Produced_Connection_Id	UINT	DeviceNet specifications	
5	Get	Consumed_Connection_Id	UINT	DeviceNet specifications	
6	Get	Initial_Comm_Characteristic	BYTE	21HEX	
7	Get	Produced_Connection_Size	UINT	0xFFFF	
8	Get	Consumed_Connection_Size	UINT	0xFFFF	
9	Get/Set	Expected Packet rate	UINT	2500msec	
12	Get/Set	Watchdog Timeout Action	USINT	1(AutoDelete)	
13	Get	Produced Connection Path Length	UINT	0	
14	Get	Produced Connection Path	ARRAY OF USINT	NULL	
15	Get	Consumed Connection Path Length	UINT	0	
16	Get	Consumed Connection Path	ARRAY OF USINT	NULL	

c. Instance : 2 (Polled I/O Message)

d. Attributes

Table 6-16. Connection Object Instance2 Attributes

Attr ID	Access Rule	Name	Data type	Value	
1	Get	State	USINT	0x00	Non-existent
				0x01	Configuring
				0x02	Waiting for Connection ID
				0x03	Established
				0x04	Timed Out
				0x05	Deferred Delete
2	Get	Instance type	USINT	0x01(I/O Message)	
3	Get	Transport Class Trigger	BYTE	82HEX(Server, Class2)	
4	Get	Produced_Connection_Id	UINT	DeviceNet specifications	
5	Get	Consumed_Connection_Id	UINT	DeviceNet specifications	
6	Get	Initial_Comm_Characteristic	BYTE	0x01	
7	Get	Produced_Connection_Size	UINT	하기 참조	
8	Get	Consumed_Connection_Size	UINT	하기참조	
9	Get/Set	Expected Packet rate	UINT	0 (msec)	
12	Get/Set	Watchdog Timeout Action	USINT	0(TimeOut)	
13	Get	Produced Connection Path Length	UINT	0x0006	
14	Get/Set	Produced Connection Path	ARRAY OF USINT	20 04 24 02 30 03	
15	Get	Consumed Connection Path Length	UINT	0x0006	
16	Get/Set	Consumed Connection Path	ARRAY OF USINT	20 04 24 07 30 03	
17	Get	Produce Inhibit Time	UINT	0	

e. Services

Table 6-17. Connection Object Services

Service Code(HEX)	Need in implementation		Name
	Class	Instance	
0E	n/a	Required	Get_Attribute_Single
10	n/a	Required	Set_Attribute_Single
05	n/a	Required	Reset

6) S-Device Super Visor Object(Class ID : 0x30)

a. Instance 0 (Class)

b. Attributes

Table 6-18. S-Device Super Visor Object Instance0 Attribute

Attr ID	Need for mounting	Access Rule	Name	Data type	Vlaue
1	Required	Get	Revision	UINT	0x0002

c. Instance 1

d. Attributes

Table 6-19. S-Device Super Visor Object Instance1 Attributes

Attr ID	Need for mounting	Access Rule	Name	Data type	Value
3	Required	Get	DeviceType	Short String	"FRC"
4	Required	Get	SEMI Standard Revision Level	Short String	"E54-0997"
5	Required	Get	Manufacturer's Name	Short String	"MKP Co., Ltd."
6	Required	Get	Manufacturer's Model Number	Short String	"MADEE MODEL"
7	Required	Get	Software Revision Level	Short String	"1.00"
8	Required	Get	Hardware Revision Level	Short String	"1.00"
9	Optional	Get	Manufacturer's Serial Number	Short String	Varies by Product Max 30 Characters
11	Required	Get	Device Status	USINT	DeviceNet specifications
12	Required	Get	Exception Status	BYTE	Expanded Method
15	Required	Get	Alarm Enable	BOOL	1=Enable
16	Required	Get	Warning Enable	BOOL	1= Enable
100	Optional	Get	Number of current units	USINT	
101	Optional	Get/Set	Number of saved units	USINT	

e. Device Status

Table 6-20. Device Status Attribute Values

Attribute Value	State
0	Undefined
1	Self Testing
2	Idle
3	Self-Test Exception
4	Executing
5	About
6	Critical Fault
7-50	Reserved by CIP
51-99	Device Specific
100-255	Vendor Specific

f. Exception Status Bit Map

Table 6-21. Exception Status Bit Map

Bit	Function
0	ALARM/device-common
1	ALARM/device-specific
2	ALARM/manufacture-specific
3	reserved – set to 0
4	WARNING/device-common
5	WARNING/device-specific
6	WARNING/manufacture-specific
7	1 == Expanded Method

g. Services

Table 6-22. S-Device Super Visor Object Services

Service Code(HEX)	Need in implementation		Name
	Class	Instance	
0E	Required	Required	Get_Attribute_Single
10	n/a	Required	Set_Attribute_Single
05	n/a	Required	Reset
06	n/a	Required	Start
07	n/a	Optional	Stop
4B	n/a	Required	Abort
4C	n/a	Required	Recover
4E	n/a	Required	Perform_Diagnostics

7) S-Analog Sensor Object(Class ID : 0x31)

- a. Instance1
- b. Attributes
- c.

Table 6-23. S-Analog Sensor Object Instance1 Attributes

Attr ID	Need for mounting	Access Rule	Name	Data type	Value	
1	Optional	Get	Number of Attributes	USINT		
2	Optional	Get	Attribute List	ARRAY OF USINT		
3	Optional	Get/Set	Data Type	USINT	0xC3 = INT 0xCA = REAL	
4	Optional	Get/Set	Data Units	UINT	0x1001 = Count 0x1007 = %	
5	Required	Get	Reading Valid	BOOL	0=Invalid, 1=Valid	
6	Required	Get	Value	INT or REAL	DeviceNet Specifications	
7	Required	Get	Status	BYTE	Details(0:Clear,1:Set)	
					Bit0	High Alarm
					Bit1	Low Alarm
					Bit2	High Warning
					Bit3	Low Warning
Bit4~7:reserved						
10	Optional	Get	Full Scale	INT or REAL (based on Attribute3)	0x6000(Default) Maximum : 100%(REAL)	
101	Optional	Get	Flow Full Scale (REAL)	REAL	Unit : SCCM	

- d. Instance Number

Depending on the number of the FRC Unit, the number of Instance varies. The maximum instance value is equal to the number of FRC units.

- e. Services

Table 6-24. S-Analog Sensor Object Services

Service Code(HEX)	Need in implementation		Name
	Class	Instance	
0E	n/a	Required	Get_Attribute_Single
10	n/a	Required	Set_Attribute_Single

8) S-Single Stage Controller Object(Class ID : 0x33)

a. Instance : 1

b. Attributes

Table 6-25. S-Single Stage Controller Object Instance1 Attributes

Attr ID	Need for mounting	Access Rule	Name	Data type	Value
1	Optional	Get	Number of Attribute	USINT	
2	Optional	Get	Attribute List	ARRAY OF USINT	
3	Optional	Get/Set	Data Type	USINT	0xC3 = INT
4	Optional	Get/Set	Data Units	UINT	0x1001 = Count
5	Optional	Get/Set	Control Mode	USINT	0X00
6	Required	Get/Set	Ratio Input	INT	하기 참조
7	Required	Get	Ratio Output	INT	하기 참조
10	Required	Get	Status	BYTE	Bit0 Alarm
					Bit1 Warning
					Bit2~7 : Reserved
17	Optional	Get/Set	Safe State	USINT	0x00(Default)
110	Optional	Get	Ratio FullScale	USINT	0x8000

c. Services

Table 6-26. S-Single Stage Controller Object Services

Service Code(HEX)	Need in implementation		Name
	Class	Instance	
0E	n/a	Required	Get_Attribute_Single
10	n/a	Required	Set_Attribute_Single

d. Ratio Input/Output Data Format

Table 6-27. Ratio Input Data Format of Total Unit N

Unit	Byte	Data
2	0	Master Unit Ratio Input / Output (high byte)
	1	Master Unit Ratio Input / Output (Low byte)
	2	Slave1 Unit Ratio Input / Output (high byte)
	3	Slave1 Unit Ratio Input / Output (Low byte)

N	0	Master Unit Ratio Input / Output (high byte)
	1	Master Unit Ratio Input / Output (Low byte)
	...	...
	...	
	2N-2	Slave N-1 Unit Ratio Input / Output (high byte)
	2N-1	Slave N-1 Unit Ratio Input / Output (Low byte)

e. Ratio Full Scale

It is a value that represents the total value of the ratio input / output used for Device Net communication.

The sum value of Ratio Input / Output is 0x8000 (32768), indicating 100%.

7. Specifications of MADEE Series

7-1. Performance Specifications

Table 7-1. Performance Specifications

Parameter	Description	
	Divide Type	Integrated Type
Flow Rate Range (N2 Equivalent)	5SCCM – 200 SLM	5SCCM – 20 SLM
Channel	2 - 6 Channel	2 - 4 Channel
Flow Ratio Control Range	0, 2 – 100%	
Flow Accuracy	within $\pm 0.5\%$ Full Scale (2~25%) within $\pm 2.0\%$ Reading Scale (25~100%)	
Response Time	within 3 sec	
Linearity	within $\pm 0.3\%$ F.S	
Repeatability	within $\pm 0.2\%$ F.S	
Leak Across Valve	within 2% F.S	
Leak Integrity	$\leq 1 \times 10^{-11}$ Pa · m ³ /sec(He)	
Operating Temperature	5 ~ 50°C (15 ~ 35°C guaranteed)	
Warm-up Time	30 min(minimum)	
Max Operating Outlet Pressure	150 Psig	
Zero Stability	within $\pm 0.5\%$ F.S per year	

7-2. Mechanical Specifications

Table 7-2. Mechanical Specifications

Parameter	Description
Control Valve Type	Normally Open, Solenoid Valve.
Seals	Metal
Material for Gas-wetted Surface	SUS316L, SUS329J1
Standard Fittings	1/2" VCR
Surface Finish for Gas-wetted Surface	Below Ra0.2

7-3. Electrical Specifications

Table 7-3. Electrical Specifications

Parameter	Description
Electrical Interface Connector	DeviceNet™ Male Micro Style Connector
Power Requirement	+24VDC more than 320mA(1 Unit)
Power Consumption	7.5W max
Input Signal	DeviceNet (125K, 250K, 500Kbps)
Output Signal	DeviceNet (125K, 250K, 500Kbps)

8. Example for in use

8-1. Switch Setup

Table 8-1. Baud Rate Switch Assign

Switch Position	Baud Rate
0	125Kbps
1	250Kbps
2	500Kbps
3~255	Program Mode

The Baud Rate Switch can be found on top of the device and be detected when the device Power-Up. If you change position of this switch during device operating, that value can't use on device until re-power-up or reset. You can get this value to use Get_Attribute_Single service on DeviceNet Object.

Table 8-2. MAC ID Switch Assign

"Tens" Position	"Ones" Position	MAC ID
0	0~9	0 ~ 9
1	0~9	10 ~ 19
2	0~9	20 ~ 29
3	0~9	30 ~ 39
4	0~9	40 ~ 49
5	0~9	50 ~ 59
6	0~3	60 ~ 63
6	4~9	Program Mode
7	0~9	Program Mode
8	0~9	Program Mode
9	0~9	Program Mode

The MAC ID Switch can be found beside the Baud Rate Switch. The Value of MAC ID Switch be refreshed after power up. You can get this value to use Get_Attribute_Single service on DeviceNet Object.

8-2. Explicit Message Process

Table 8-3.Group2 Identifier Field

Bit	10	9	8	7	6	5	4	3	2	1	0
Mean	1	0	MACID						Group2 MessageID		

Table 8-4.Non-Fragmented Explicit Request Message Body Field

	7	6	5	4	3	2	1	0	
0Byte	Flag [0]	XID	MAC ID						Message Header
1	R/R [0]	Service Code						Message Body	
2	Class ID								
3	Instance ID								
4	Attribute ID								

Message Body isn't always same. Be changed by service and sending byte size.

a. Group2_UCMM_Allocate(EM Open)

Req/Res	Field	Hex	Description
Req	Identifier	04 0E	Group2, DestMac 1, MessageID 6
	Data	00 4B 03 01 01 00	Group2 UCMM Allocate Request
Res	Identifier	04 0B	Group2, DestMac SourceMac 1, MessageID3
	Data	00 CB 00	Group2 UCMM Allocate Response

b. Get_Attribute_Single Service

Req/Res	Field	Hex	Description
Req	Identifier	04 0C	Group2, DestMac 1, MessageID 4
	Data	00 0E 01 01 01	Get Single Service, Identity Object, Vendor ID
Res	Identifier	04 0B	Group2, DestMac SourceMac 1, MessageID3
	Data	00 8E BD 04	Vendor ID = 0x04BD (1213 dec)

c. Set_Attribute_Single Service

Req/Res	Field	Hex	Description
Req	Identifier	04 0C	Group2, DestMac 1, MessageID 4
	Data	40 10 05 02 09 C8 00	Set Single Service, Connection Object, Instace02, Expected Packet Rate 200msec
Res	Identifier	04 0B	Group2, DestMac SourceMac 1, MessageID3
	Data	40 90 C8 00	Expected Packet Rate 200msec

8-3. Polled I/O Message Process

Table 8-4. Polled Predefined Master/Slave Identifier Field

Bit	10	9	8	7	6	5	4	3	2	1	0	Description
Mean	0	Group 1 Message ID				Source MAC ID						Group 1 Message
	0	1	1	1	1	Slave's MAC ID						Slave's I/O Poll Response Message
	1	0	MAC ID						Group2 MessageID			Group 3 Message
	1	0	Slave's MAC ID						1	0	1	Master's I/O Poll Command Message

Master To/From MAC ID 09

Poll Command Message
Master's MAC ID = 01
Slaves's MAC ID = 09
CAN Identifier Field = 44D_{hex}

Poll Response Message
Master's MAC ID = 01
Slaves's MAC ID = 09
CAN Identifier Field = 3C9_{hex}

Group 2 Message
Destination MAC ID = 09
Group 2 Message ID = 5
Poll Output Data (0 - 8 bytes)
Identifier = 10 001001 101, Data = 0102030405060708

Group 1 Message
Group 1 Message ID = F
Source MAC ID = 09
Poll Response Data (0 - 8 bytes)
Identifier = 0 1111 001001, Data = 0102030405060708

Master To/From MAC ID 11

Poll Command Message
Master's MAC ID = 01
Slaves's MAC ID = 11
CAN Identifier Field = 45D_{hex}

Poll Response Message
Master's MAC ID = 01
Slaves's MAC ID = 11
CAN Identifier Field = 3CB_{hex}

Group 2 Message
Destination MAC ID = 11 (0B hex)
Group 2 Message ID = 5
Poll Output Data of 1 byte, can be 0 - 8 bytes
Identifier = 10 001011 101, Data = 01

Group 1 Message
Group 1 Message ID = F
Source MAC ID = 11 (0B hex)
Poll Response Data of 2 bytes, can be 0 - 8 bytes
Identifier = 0 1111 001011, Data = 0102



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